

```
import pandas as pd
from scipy.stats import ttest_ind
import matplotlib.pyplot as plt
```

```
data = pd.read_csv("QVI_data.csv")
```

data

	LYLTY_CARD_NBR	DATE	STORE_NBR	TXN_ID	PROD_NBR	PROD_NAME	PF
0	1000	2018-10-17	1	1	5	Natural Chip Compny SeaSalt175g	
1	1002	2018-09-16	1	2	58	Red Rock Deli Chikn&Garlic Aioli 150g	
2	1003	2019-03-07	1	3	52	Grain Waves Sour Cream&Chives 210G	
3	1003	2019-03-08	1	4	106	Natural ChipCo Hony Soy Chckn175g	
4	1004	2018-11-02	1	5	96	WW Original Stacked Chips 160g	
...	...	...	...	...	...	...	...
264829	2370701	2018-12-08	88	240378	24	Grain Waves Sweet Chilli 210g	
264830	2370751	2018-10-01	88	240394	60	Kettle Tortilla ChpsFeta&Garlic 150g	
264831	2370961	2018-10-24	88	240480	70	Tyrrells Crisps Lightly Salted 165g	
264832	2370961	2018-10-27	88	240481	65	Old El Paso Salsa Dip Chnky Tom Ht300g	
264833	2373711	2018-12-14	88	241815	16	Smiths Crinkle Chips Salt & Vinegar 330g	

264834 rows × 12 columns

```
data.columns
```

```
Index(['LYLTY_CARD_NBR', 'DATE', 'STORE_NBR', 'TXN_ID', 'PROD_NBR',  
      'PROD_NAME', 'PROD_QTY', 'TOT_SALES', 'PACK_SIZE', 'BRAND',  
      'LIFESTAGE',  
      'PREMIUM_CUSTOMER'],  
      dtype='object')
```

```
#Converting the DATE type(object) to type datetime  
data["DATE"] = pd.to_datetime(data["DATE"])
```

```
print(data['DATE'].dtype)
```

```
datetime64[ns]
```

```
#creating an column (year_month)  
data['YEAR_MONTH'] = data['DATE'].dt.to_period('M')
```

```
data[['DATE', 'YEAR_MONTH']].head()
```

	DATE	YEAR_MONTH
0	2018-10-17	2018-10
1	2018-09-16	2018-09
2	2019-03-07	2019-03
3	2019-03-08	2019-03
4	2018-11-02	2018-11

```
#creating a monthly store metrics for each stores  
monthly_metrics = (  
    data.groupby(["STORE_NBR", "YEAR_MONTH"])  
        .agg(  
            TOTAL_SALES=("TOT_SALES", "sum"),  
            TOTAL_CUSTOMERS=("LYLTY_CARD_NBR", "nunique"),  
            TOTAL_TXNS=("TXN_ID", "nunique")  
        )  
    .reset_index()  
)
```

```
#adding average transactions per customer  
monthly_metrics["TXN_PER_CUSTOMER"] = (  
    monthly_metrics["TOTAL_TXNS"] /  
    monthly_metrics["TOTAL_CUSTOMERS"]  
)
```

```
monthly_metrics
```

	STORE_NBR	YEAR_MONTH	TOTAL_SALES	TOTAL_CUSTOMERS	TOTAL_TXNS	TXN_PE
0	1	2018-07	206.9	49	52	
1	1	2018-08	176.1	42	43	
2	1	2018-09	278.8	59	62	
3	1	2018-10	188.1	44	45	
4	1	2018-11	192.6	46	47	
...	...	...	...	...	...	...
3164	272	2019-02	395.5	45	48	
3165	272	2019-03	442.3	50	53	
3166	272	2019-04	445.1	54	55	
3167	272	2019-05	314.6	34	40	
3168	272	2019-06	312.1	34	37	

3169 rows × 6 columns

Next steps:

[Generate code with monthly_metrics](#)

[New interactive sheet](#)

```
monthly_metrics["YEAR_MONTH"].unique()
```

```
<PeriodArray>
['2018-07', '2018-08', '2018-09', '2018-10', '2018-11', '2018-12', '2019-01',
 '2019-02', '2019-03', '2019-04', '2019-05', '2019-06']
Length: 12, dtype: period[M]
```

```
pre_trial = monthly_metrics[
    monthly_metrics["YEAR_MONTH"] < "2019-02"
]
```

```
'''pre_trial = monthly_metrics[
    monthly_metrics["YEAR_MONTH"] < pd.Period("2019-02")
]'''
```

```
'pre_trial = monthly_metrics[\n    monthly_metrics["YEAR_MONTH"] < pd.Period("2019-02")\n]'
```

```
#focusing on trial stores only
trial_stores = [77, 86, 88]

pre_trial[
    pre_trial["STORE_NBR"].isin(trial_stores)
].head()
```

	STORE_NBR	YEAR_MONTH	TOTAL_SALES	TOTAL_CUSTOMERS	TOTAL_TXNS	TXN_PER
	880	77	2018-07	296.8	51	55
	881	77	2018-08	255.5	47	48
	882	77	2018-09	225.2	42	44
	883	77	2018-10	204.5	37	38
	884	77	2018-11	245.3	41	44

```
pre_trial.groupby("STORE_NBR")["YEAR_MONTH"].count().sort_values()
```

	YEAR_MONTH
STORE_NBR	
211	1
206	1
85	1
193	1
76	1
...	...
269	7
270	7
271	7
272	7
1	7

271 rows × 1 columns

dtype: int64

```
#Choosing only that stores which are eligible
store_counts = (
    pre_trial.groupby("STORE_NBR")["YEAR_MONTH"]
    .count()
)

eligible_stores = store_counts[
    store_counts == 7
].index

pre_trial = pre_trial[
    pre_trial["STORE_NBR"].isin(eligible_stores)
]
```

```
# Correlation function
def calculate_correlation(data, trial_store, metric):
    """
    Calculate Pearson correlation between a trial store
    and every other store for a given metric.
    """
    # Trial store data
    trial = data[data["STORE_NBR"] == trial_store][
        ["YEAR_MONTH", metric]
    ].rename(columns={metric: "TRIAL_METRIC"})

    correlations = []
    for store in data["STORE_NBR"].unique():

        if store == trial_store:
            continue
        control = data[data["STORE_NBR"] == store][
            ["YEAR_MONTH", metric]
        ].rename(columns={metric: "CONTROL_METRIC"})
        merged = pd.merge(
            trial,
            control,
            on="YEAR_MONTH",
            how="inner"
        )
        if len(merged) < 2:
            continue
        corr = merged["TRIAL_METRIC"].corr(
            merged["CONTROL_METRIC"]
        )
        correlations.append({
            "Trial_Store": trial_store,
            "Control_Store": store,
            "Metric": metric,
            "Correlation": corr
        })
    return pd.DataFrame(correlations)
```

```
sales_corr = calculate_correlation(
    pre_trial,
    77,
    "TOTAL_SALES"
)

sales_corr.sort_values(
    "Correlation",
    ascending=False
).head(10)
```

	Trial_Store	Control_Store	Metric	Correlation	
67	77	71	TOTAL_SALES	0.914106	
220	77	233	TOTAL_SALES	0.903774	
110	77	119	TOTAL_SALES	0.867664	
15	77	17	TOTAL_SALES	0.842668	
2	77	3	TOTAL_SALES	0.806644	
38	77	41	TOTAL_SALES	0.783232	
46	77	50	TOTAL_SALES	0.763866	
148	77	157	TOTAL_SALES	0.735893	
153	77	162	TOTAL_SALES	0.729740	
243	77	257	TOTAL_SALES	0.724927	

```
customer_corr = calculate_correlation(
    pre_trial,
    77,
    "TOTAL_CUSTOMERS"
)

customer_corr.sort_values(
    "Correlation",
    ascending=False
).head()
```

	Trial_Store	Control_Store	Metric	Correlation	
220	77	233	TOTAL_CUSTOMERS	0.990358	
110	77	119	TOTAL_CUSTOMERS	0.983267	
240	77	254	TOTAL_CUSTOMERS	0.916208	
105	77	113	TOTAL_CUSTOMERS	0.901348	
78	77	84	TOTAL_CUSTOMERS	0.858571	

```
def calculate_magnitude(data, trial_store, metric):

    trial = data[data["STORE_NBR"] == trial_store][
        ["YEAR_MONTH", metric]
    ].rename(columns={metric: "TRIAL_METRIC"})

    similarity = []

    for store in data["STORE_NBR"].unique():

        if store == trial_store:
```

```
        continue

    control = data[data["STORE_NBR"] == store][
        ["YEAR_MONTH", metric]
    ].rename(columns={metric: "CONTROL_METRIC"})

    merged = pd.merge(
        trial,
        control,
        on="YEAR_MONTH"
    )

    if len(merged) < 2:
        continue

    distance = (
        abs(
            merged["TRIAL_METRIC"] -
            merged["CONTROL_METRIC"]
        )
    ).mean()

    similarity.append({
        "Trial_Store": trial_store,
        "Control_Store": store,
        "Distance": distance
    })

similarity = pd.DataFrame(similarity)

min_dist = similarity["Distance"].min()
max_dist = similarity["Distance"].max()

similarity["Magnitude"] = (
    1 -
    (
        similarity["Distance"] - min_dist
    ) /
    (
        max_dist - min_dist
    )
)

return similarity
```

```
sales_mag = calculate_magnitude(
    pre_trial,
    77,
    "TOTAL_SALES"
)

sales_mag.sort_values(
    "Magnitude",
```

```
ascending=False
).head()
```

	Trial_Store	Control_Store	Distance	Magnitude	
220	77	233	18.828571	1.000000	
241	77	255	28.571429	0.991889	
179	77	188	31.100000	0.989784	
49	77	53	32.471429	0.988642	
122	77	131	33.042857	0.988167	

```
comparison = pd.merge(
    sales_corr[
        ["Control_Store", "Correlation"]
    ],
    sales_mag[
        ["Control_Store", "Magnitude"]
    ],
    on="Control_Store"
)
```

```
comparison["Final_Score"] = (
    comparison["Correlation"] +
    comparison["Magnitude"]
) / 2
```

```
comparison.sort_values(
    "Final_Score",
    ascending=False
).head(10)
```

	Control_Store	Correlation	Magnitude	Final_Score	
220	233	0.903774	1.000000	0.951887	
38	41	0.783232	0.977927	0.880579	
46	50	0.763866	0.986775	0.875320	
15	17	0.842668	0.886125	0.864397	
107	115	0.689159	0.942700	0.815929	
158	167	0.657110	0.971148	0.814129	
251	265	0.639759	0.975489	0.807624	
221	234	0.696325	0.896056	0.796190	
49	53	0.532764	0.988642	0.760703	
78	84	0.684348	0.832583	0.758465	


```
# for 77 the control store store is 233
```

```
sales_corr = calculate_correlation(  
    pre_trial,  
    86,  
    "TOTAL_SALES"  
)  
  
sales_corr.sort_values(  
    "Correlation",  
    ascending=False  
) .head(10)
```

	Trial_Store	Control_Store	Metric	Correlation	
146	86	155	TOTAL_SALES	0.877882	
123	86	132	TOTAL_SALES	0.846517	
227	86	240	TOTAL_SALES	0.825066	
209	86	222	TOTAL_SALES	0.795075	
101	86	109	TOTAL_SALES	0.788300	
129	86	138	TOTAL_SALES	0.759864	
188	86	198	TOTAL_SALES	0.748794	
106	86	114	TOTAL_SALES	0.734415	
20	86	22	TOTAL_SALES	0.722966	
5	86	6	TOTAL_SALES	0.721892	

```
customer_corr = calculate_correlation(  
    pre_trial,  
    86,  
    "TOTAL_CUSTOMERS"  
)  
  
customer_corr.sort_values(  
    "Correlation",  
    ascending=False  
) .head()
```

	Trial_Store	Control_Store	Metric	Correlation	
146	86	155	TOTAL_CUSTOMERS	0.942876	
106	86	114	TOTAL_CUSTOMERS	0.855339	
246	86	260	TOTAL_CUSTOMERS	0.846502	
167	86	176	TOTAL_CUSTOMERS	0.796380	
101	86	109	TOTAL_CUSTOMERS	0.770778	

```

sales_mag = calculate_magnitude(
    pre_trial,
    86,
    "TOTAL_SALES"
)

sales_mag.sort_values(
    "Magnitude",
    ascending=False
).head()

```

	Trial_Store	Control_Store	Distance	Magnitude	
101	86	109	31.607143	1.000000	
146	86	155	32.464286	0.998964	
209	86	222	35.864286	0.994853	
212	86	225	37.535714	0.992833	
216	86	229	45.864286	0.982764	

```

comparison = pd.merge(
    sales_corr[
        ["Control_Store", "Correlation"]
    ],
    sales_mag[
        ["Control_Store", "Magnitude"]
    ],
    on="Control_Store"
)

comparison["Final_Score"] = (
    comparison["Correlation"] +
    comparison["Magnitude"]
) / 2

comparison.sort_values(
    "Final_Score",
    ascending=False
).head(10)

```

	Control_Store	Correlation	Magnitude	Final_Score	
146	155	0.877882	0.998964	0.938423	
209	222	0.795075	0.994853	0.894964	
101	109	0.788300	1.000000	0.894150	
129	138	0.759864	0.959103	0.859483	
106	114	0.734415	0.955087	0.844751	
255	269	0.697055	0.934267	0.815661	
98	106	0.678684	0.943602	0.811143	
212	225	0.617533	0.992833	0.805183	
119	128	0.654929	0.895719	0.775324	
74	79	0.668748	0.861479	0.765113	

86 the control store is 155

```

sales_corr = calculate_correlation(
    pre_trial,
    88,
    "TOTAL_SALES"
)

sales_corr.sort_values(
    "Correlation",
    ascending=False
).head(10)

```

	Trial_Store	Control_Store	Metric	Correlation	
150	88	159	TOTAL_SALES	0.903186	
194	88	204	TOTAL_SALES	0.885774	
125	88	134	TOTAL_SALES	0.864293	
0	88	1	TOTAL_SALES	0.813636	
239	88	253	TOTAL_SALES	0.811838	
84	88	91	TOTAL_SALES	0.776688	
57	88	61	TOTAL_SALES	0.748929	
169	88	178	TOTAL_SALES	0.731857	
179	88	188	TOTAL_SALES	0.716752	
152	88	161	TOTAL_SALES	0.706581	

```
customer_corr = calculate_correlation(
    pre_trial,
    88,
    "TOTAL_CUSTOMERS"
)

customer_corr.sort_values(
    "Correlation",
    ascending=False
).head()
```

	Trial_Store	Control_Store	Metric	Correlation	
224	88	237	TOTAL_CUSTOMERS	0.947326	
12	88	14	TOTAL_CUSTOMERS	0.942976	
169	88	178	TOTAL_CUSTOMERS	0.939466	
32	88	35	TOTAL_CUSTOMERS	0.899594	
105	88	113	TOTAL_CUSTOMERS	0.862632	

```
sales_mag = calculate_magnitude(
    pre_trial,
    88,
    "TOTAL_SALES"
)

sales_mag.sort_values(
    "Magnitude",
    ascending=False
).head()
```

	Trial_Store	Control_Store	Distance	Magnitude	
224	88	237	58.428571	1.000000	
193	88	203	66.100000	0.993943	
37	88	40	83.028571	0.980578	
189	88	199	104.457143	0.963659	
156	88	165	113.057143	0.956869	

```

comparison = pd.merge(
    sales_corr[
        ["Control_Store", "Correlation"]
    ],
    sales_mag[
        ["Control_Store", "Magnitude"]
    ],
    on="Control_Store"
)

comparison["Final_Score"] = (
    comparison["Correlation"] +
    comparison["Magnitude"]
) / 2

comparison.sort_values(
    "Final_Score",
    ascending=False
).head(10)

```

	Control_Store	Correlation	Magnitude	Final_Score	
116	125	0.624109	0.890950	0.757529	
193	203	0.508001	0.993943	0.750972	
84	91	0.776688	0.690439	0.733564	
6	7	0.649657	0.812646	0.731151	
169	178	0.731857	0.725911	0.728884	
191	201	0.492735	0.912115	0.702425	
98	106	0.644724	0.730643	0.687683	
74	79	0.576960	0.793432	0.685196	
224	237	0.308479	1.000000	0.654240	
146	155	0.600771	0.699315	0.650043	

for 88 the control store is 125

```
comparison.sort_values(
    "Final_Score",
    ascending=False
).head(10)
```

	Control_Store	Correlation	Magnitude	Final_Score	
116	125	0.624109	0.890950	0.757529	
193	203	0.508001	0.993943	0.750972	
84	91	0.776688	0.690439	0.733564	
6	7	0.649657	0.812646	0.731151	
169	178	0.731857	0.725911	0.728884	
191	201	0.492735	0.912115	0.702425	
98	106	0.644724	0.730643	0.687683	
74	79	0.576960	0.793432	0.685196	
224	237	0.308479	1.000000	0.654240	
146	155	0.600771	0.699315	0.650043	

```
#trail period data
trial_period = monthly_metrics[
    (monthly_metrics["YEAR_MONTH"] >= pd.Period("2019-02")) &
    (monthly_metrics["YEAR_MONTH"] <= pd.Period("2019-04"))
]
```

```
# trail store vs control store comparsion for 77 and 233
trial_77 = monthly_metrics[
    monthly_metrics["STORE_NBR"] == 77
]

control_233 = monthly_metrics[
    monthly_metrics["STORE_NBR"] == 233
]
compare_77 = pd.merge(
    trial_77,
    control_233,
    on="YEAR_MONTH",
    suffixes=("_TRIAL", "_CONTROL")
)

compare_77
```

	STORE_NBR_TRIAL	YEAR_MONTH	TOTAL_SALES_TRIAL	TOTAL_CUSTOMERS_TRIAL	TOTAL_CUSTOMERS_CONTROL
0	77	2018-07	296.8	51	47
1	77	2018-08	255.5	47	42
2	77	2018-09	225.2	42	37
3	77	2018-10	204.5	37	41
4	77	2018-11	245.3	41	46
5	77	2018-12	267.3	46	35
6	77	2019-01	204.4	35	45
7	77	2019-02	235.0	45	50
8	77	2019-03	278.5	50	47
9	77	2019-04	263.5	47	55
10	77	2019-05	299.3	55	41
11	77	2019-06	264.7	41	

Next steps:

[Generate code with compare_77](#)[New interactive sheet](#)Start coding or [generate](#) with AI.

```
# trail store vs control store comparsion for 86 and 155
trial_86 = monthly_metrics[
    monthly_metrics["STORE_NBR"] == 86
]

control_155 = monthly_metrics[
    monthly_metrics["STORE_NBR"] == 155
]

compare_86 = pd.merge(
    trial_86,
    control_155,
    on="YEAR_MONTH",
    suffixes=("_TRIAL", "_CONTROL")
)

compare_86
```

	STORE_NBR_TRIAL	YEAR_MONTH	TOTAL_SALES_TRIAL	TOTAL_CUSTOMERS_TRIAL	TOTAL_CUSTOMERS_CONTROL
0	86	2018-07	892.20	99	99
1	86	2018-08	764.05	94	94
2	86	2018-09	914.60	103	103
3	86	2018-10	948.40	109	109
4	86	2018-11	918.00	100	100
5	86	2018-12	841.20	98	98
6	86	2019-01	841.40	94	94
7	86	2019-02	913.20	107	107
8	86	2019-03	1026.80	115	115
9	86	2019-04	848.20	105	105
10	86	2019-05	889.30	104	104
11	86	2019-06	838.00	98	98

Next steps:

[Generate code with compare_86](#)[New interactive sheet](#)Start coding or [generate](#) with AI.

```
# trail store vs control store comparsion for 88 and 125
trial_88 = monthly_metrics[
    monthly_metrics["STORE_NBR"] == 88
]

control_125 = monthly_metrics[
    monthly_metrics["STORE_NBR"] == 125
]
compare_88 = pd.merge(
    trial_88,
    control_125,
    on="YEAR_MONTH",
    suffixes=("_TRIAL", "_CONTROL")
)

compare_88
```


	STORE_NBR_TRIAL	YEAR_MONTH	TOTAL_SALES_TRIAL	TOTAL_CUSTOMERS_TRIAL	TOTAL_CUSTOMERS_CONTROL
0	88	2018-07	1310.00	129	129
1	88	2018-08	1323.80	131	131
2	88	2018-09	1423.00	124	124
3	88	2018-10	1352.40	123	123
4	88	2018-11	1382.80	130	130
5	88	2018-12	1325.20	126	126
6	88	2019-01	1266.40	117	117
7	88	2019-02	1370.20	124	124
8	88	2019-03	1477.20	134	134
9	88	2019-04	1439.40	128	128
10	88	2019-05	1308.25	128	128
11	88	2019-06	1354.60	121	121

Next steps:

[Generate code with compare_88](#)[New interactive sheet](#)

```

#Statistical Test for 77

# for sales
trial_sales = compare_77[
    compare_77["YEAR_MONTH"] >= pd.Period("2019-02")
]["TOTAL_SALES_TRIAL"]

control_sales = compare_77[
    compare_77["YEAR_MONTH"] >= pd.Period("2019-02")
]["TOTAL_SALES_CONTROL"]

t_stat, p_value = ttest_ind(
    trial_sales,
    control_sales
)

print('T test for trail sales :',t_stat)
print('p value for control sales :',p_value)
print()

#for customers
trial_customers = compare_77[
    compare_77["YEAR_MONTH"] >= pd.Period("2019-02")
]["TOTAL_CUSTOMERS_TRIAL"]

control_customers = compare_77[
    compare_77["YEAR_MONTH"] >= pd.Period("2019-02")
]

```

```
]["TOTAL_CUSTOMERS_CONTROL"]

t_stat, p_value = ttest_ind(
    trial_customers,
    control_customers
)

print('T test for trail customers :',t_stat)
print('p value for control customers :',p_value)
print()

#for transactions per customers
trial_transaction = compare_77[
    compare_77["YEAR_MONTH"] >= pd.Period("2019-02")
]["TXN_PER_CUSTOMER_TRIAL"]

control_transaction = compare_77[
    compare_77["YEAR_MONTH"] >= pd.Period("2019-02")
]["TXN_PER_CUSTOMER_CONTROL"]

t_stat, p_value = ttest_ind(
    trial_transaction,
    control_transaction
)

print('T test for trail transaction per customers :',t_stat)
print('p value for control transaction per customers :',p_value)
```

T test for trail sales : 1.0590935477424306
p value for control sales : 0.32048413679130267

T test for trail customers : 1.0076877522384466
p value for control customers : 0.3431073698911317

T test for trail transaction per customers : -0.5179629957257171
p value for control transaction per customers : 0.6184997395977818

Start coding or [generate](#) with AI.

Next steps:

[Generate code with compare_77](#)[New interactive sheet](#)

```
#Statistical Test for 86

# for sales
trial_sales = compare_86[
    compare_86["YEAR_MONTH"] >= pd.Period("2019-02")
]["TOTAL_SALES_TRIAL"]

control_sales = compare_86[
    compare_86["YEAR_MONTH"] >= pd.Period("2019-02")
]["TOTAL_SALES_CONTROL"]

t_stat, p_value = ttest_ind(
    trial_sales,
    control_sales
)

print('T test for trail sales :',t_stat)
print('p value for control sales :',p_value)
print()

#for customers
trial_customers = compare_86[
    compare_86["YEAR_MONTH"] >= pd.Period("2019-02")
]["TOTAL_CUSTOMERS_TRIAL"]

control_customers = compare_86[
    compare_86["YEAR_MONTH"] >= pd.Period("2019-02")
```

```
][ "TOTAL_CUSTOMERS_CONTROL" ]

t_stat, p_value = ttest_ind(
    trial_customers,
    control_customers
)

print('T test for trail customers :',t_stat)
print('p value for control customers :',p_value)
print()

#for transactions per customers
trial_transaction = compare_86[
    compare_86["YEAR_MONTH"] >= pd.Period("2019-02")
][ "TXN_PER_CUSTOMER_TRIAL" ]

control_transaction = compare_86[
    compare_86["YEAR_MONTH"] >= pd.Period("2019-02")
][ "TXN_PER_CUSTOMER_CONTROL" ]

t_stat, p_value = ttest_ind(
    trial_transaction,
    control_transaction
)

print('T test for trail transaction per customers :',t_stat)
print('p value for control transaction per customers :',p_value)
```

T test for trail sales : 0.9914861460518972
p value for control sales : 0.35048565098844203

T test for trail customers : 2.264554068289191
p value for control customers : 0.05334189534187311

T test for trail transaction per customers : -1.6799400467273102
p value for control transaction per customers : 0.13148054286474384

Start coding or [generate](#) with AI.

```
#Statistical Test for 88

# for sales
trial_sales = compare_88[
    compare_88["YEAR_MONTH"] >= pd.Period("2019-02")
][ "TOTAL_SALES_TRIAL" ]

control_sales = compare_88[
    compare_88["YEAR_MONTH"] >= pd.Period("2019-02")
][ "TOTAL_SALES_CONTROL" ]

t_stat, p_value = ttest_ind(
    trial_sales,
    control_sales
```

```
)

print('T test for trail sales :',t_stat)
print('p value for control sales :',p_value)
print()

#for customers
trial_customers = compare_88[
    compare_88["YEAR_MONTH"] >= pd.Period("2019-02")
]["TOTAL_CUSTOMERS_TRIAL"]

control_customers = compare_88[
    compare_88["YEAR_MONTH"] >= pd.Period("2019-02")
]["TOTAL_CUSTOMERS_CONTROL"]

t_stat, p_value = ttest_ind(
    trial_customers,
    control_customers
)

print('T test for trail customers :',t_stat)
print('p value for control customers :',p_value)
print()

#for transactions per customers
trial_transaction = compare_88[
    compare_88["YEAR_MONTH"] >= pd.Period("2019-02")
]["TXN_PER_CUSTOMER_TRIAL"]

control_transaction = compare_88[
    compare_88["YEAR_MONTH"] >= pd.Period("2019-02")
]["TXN_PER_CUSTOMER_CONTROL"]

t_stat, p_value = ttest_ind(
    trial_transaction,
    control_transaction
)

print('T test for trail transaction per customers :',t_stat)
print('p value for control transaction per customers :',p_value)
```

```
T test for trail sales : 5.265409059030732
p value for control sales : 0.0007595067310369676
```

```
T test for trail customers : 3.701571849795534
p value for control customers : 0.006028960819866707
```

```
T test for trail transaction per customers : 2.8968022378590024
p value for control transaction per customers : 0.019989488250877888
```

```
#graphs
#sales graph for store 77
plt.figure(figsize=(10,5))
```

```

plt.plot(
    compare_77["YEAR_MONTH"].astype(str),
    compare_77["TOTAL_SALES_TRIAL"],
    marker="o",
    label="Trial Store 77"
)

plt.plot(
    compare_77["YEAR_MONTH"].astype(str),
    compare_77["TOTAL_SALES_CONTROL"],
    marker="o",
    label="Control Store 233"
)

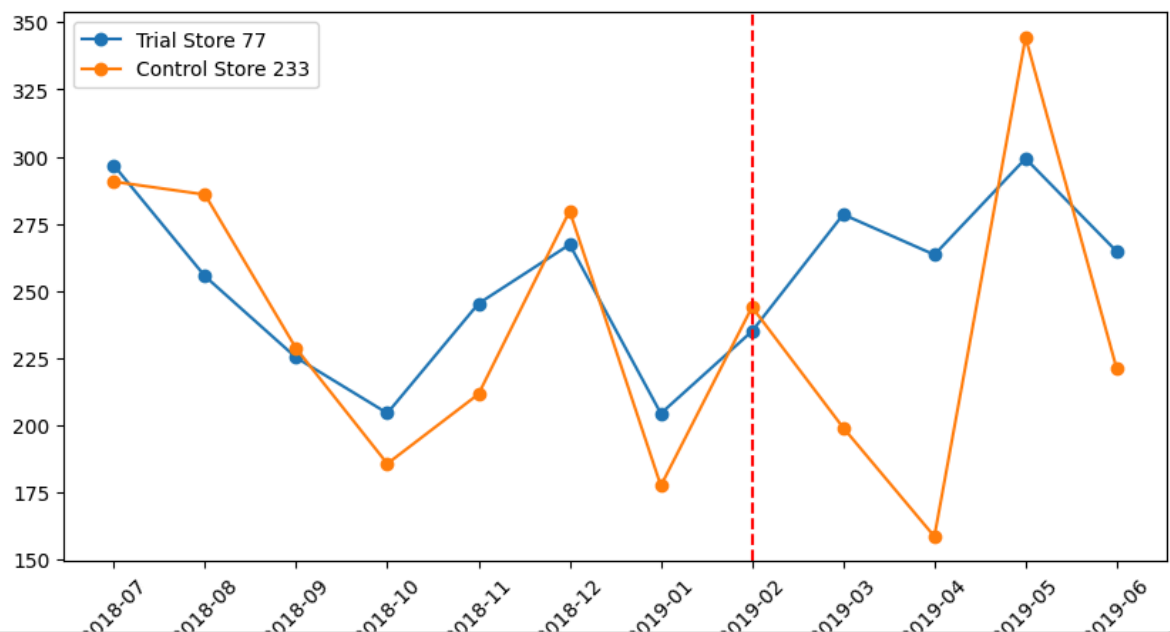
plt.axvline("2019-02",color="red",linestyle="--")

plt.legend()

plt.xticks(rotation=45)

plt.show()

```



#Trial Store 77 experienced higher sales during the trial period compared to

```

#graphs
#customers graph for store 77
plt.figure(figsize=(10,5))

plt.plot(
    compare_77["YEAR_MONTH"].astype(str),
    compare_77["TOTAL_CUSTOMERS_TRIAL"],
    marker="o",
    label="Trial Store 77"
)

```

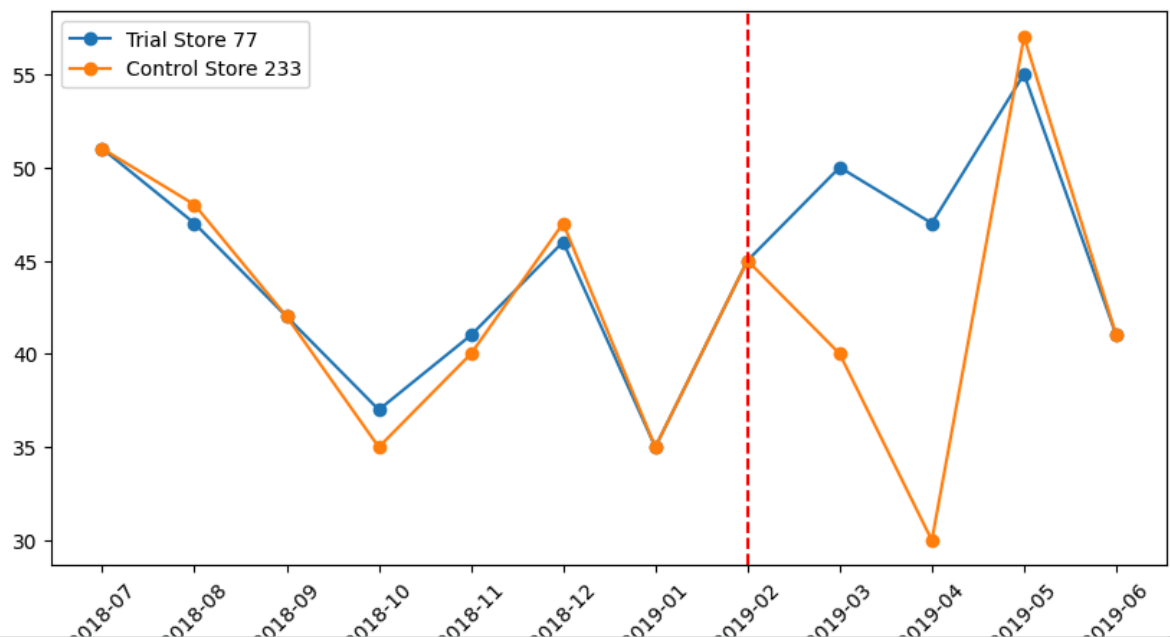
```
plt.plot(
    compare_77["YEAR_MONTH"].astype(str),
    compare_77["TOTAL_CUSTOMERS_CONTROL"],
    marker="o",
    label="Control Store 233"
)

plt.axvline("2019-02",color="red",linestyle="--")

plt.legend()

plt.xticks(rotation=45)

plt.show()
```



#The increase in sales appears to be driven by an increase in customer count

```
#graphs
#transaction per customers graph for store 77
plt.figure(figsize=(10,5))

plt.plot(
    compare_77["YEAR_MONTH"].astype(str),
    compare_77["TXN_PER_CUSTOMER_TRIAL"],
    marker="o",
    label="Trial Store 77"
)

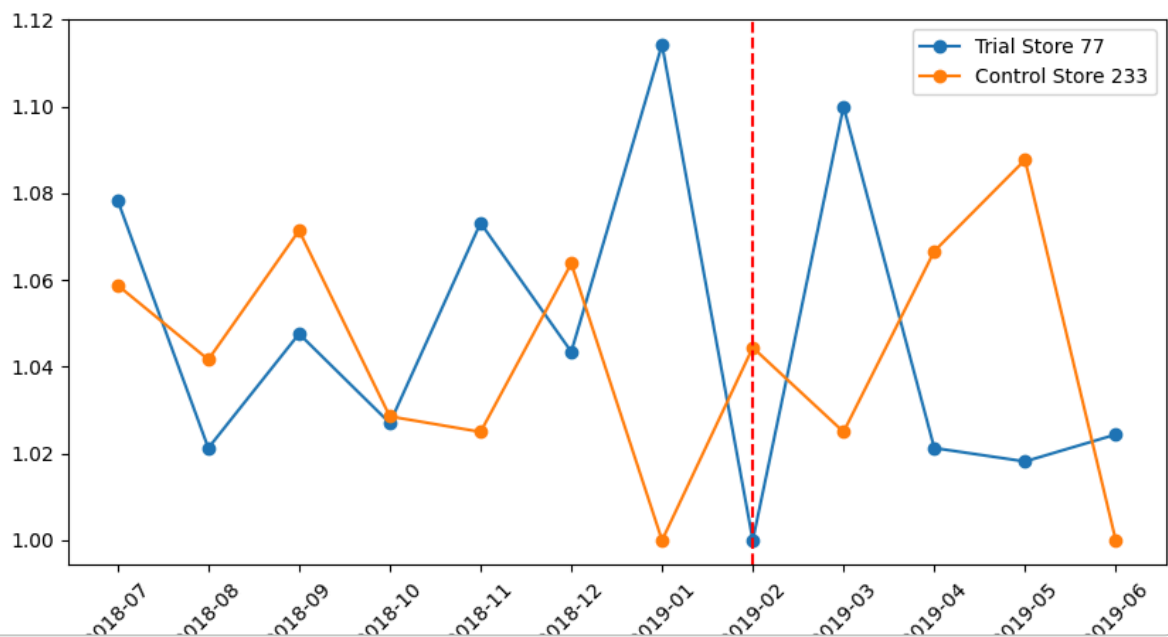
plt.plot(
    compare_77["YEAR_MONTH"].astype(str),
    compare_77["TXN_PER_CUSTOMER_CONTROL"],
    marker="o",
    label="Control Store 233"
)

plt.axvline("2019-02",color="red",linestyle="--")
```

```
plt.legend()

plt.xticks(rotation=45)

plt.show()
```



#The increase was driven by higher purchase frequency.

```
#graphs
#sales graph for store 86
plt.figure(figsize=(10,5))

plt.plot(
    compare_86["YEAR_MONTH"].astype(str),
    compare_86["TOTAL_SALES_TRIAL"],
    marker="o",
    label="Trial Store 86"
)

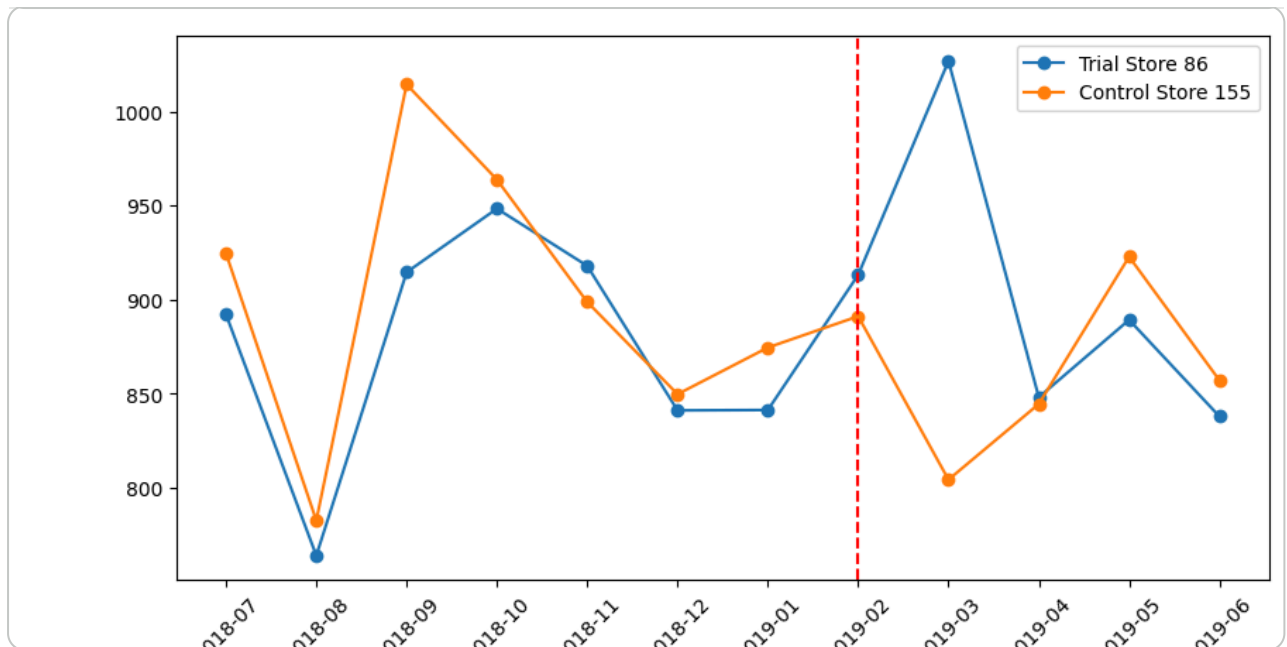
plt.plot(
    compare_86["YEAR_MONTH"].astype(str),
    compare_86["TOTAL_SALES_CONTROL"],
    marker="o",
    label="Control Store 155"
)

plt.axvline("2019-02",color="red",linestyle="--")

plt.legend()

plt.xticks(rotation=45)

plt.show()
```

#Trial Store 86 experienced higher sales during the trial period compared to

```
#graphs
#customers graph for store 86
plt.figure(figsize=(10,5))

plt.plot(
    compare_86["YEAR_MONTH"].astype(str),
    compare_86["TOTAL_CUSTOMERS_TRIAL"],
    marker="o",
    label="Trial Store 86"
)

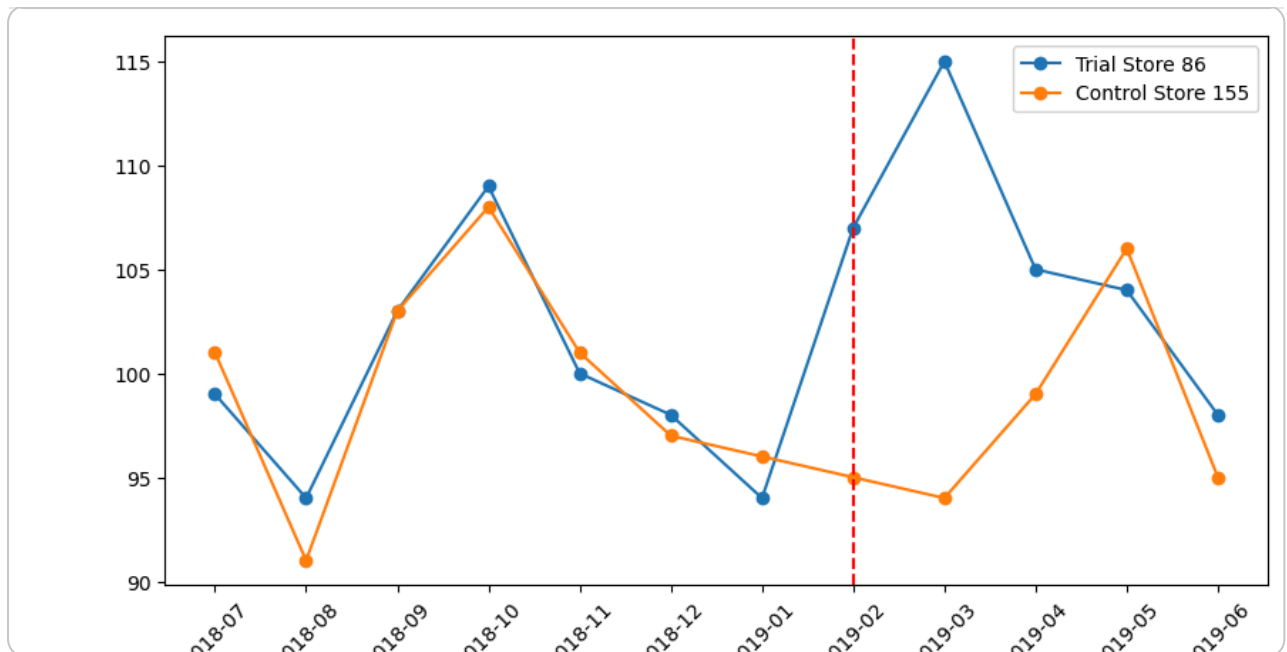
plt.plot(
    compare_86["YEAR_MONTH"].astype(str),
    compare_86["TOTAL_CUSTOMERS_CONTROL"],
    marker="o",
    label="Control Store 155"
)

plt.axvline("2019-02",color="red",linestyle="--")

plt.legend()

plt.xticks(rotation=45)

plt.show()
```



#The increase in sales appears to be driven by an increase in customer count

```
#graphs
#transaction per customers graph for store 86
plt.figure(figsize=(10,5))

plt.plot(
    compare_86["YEAR_MONTH"].astype(str),
    compare_86["TXN_PER_CUSTOMER_TRIAL"],
    marker="o",
    label="Trial Store 86"
)

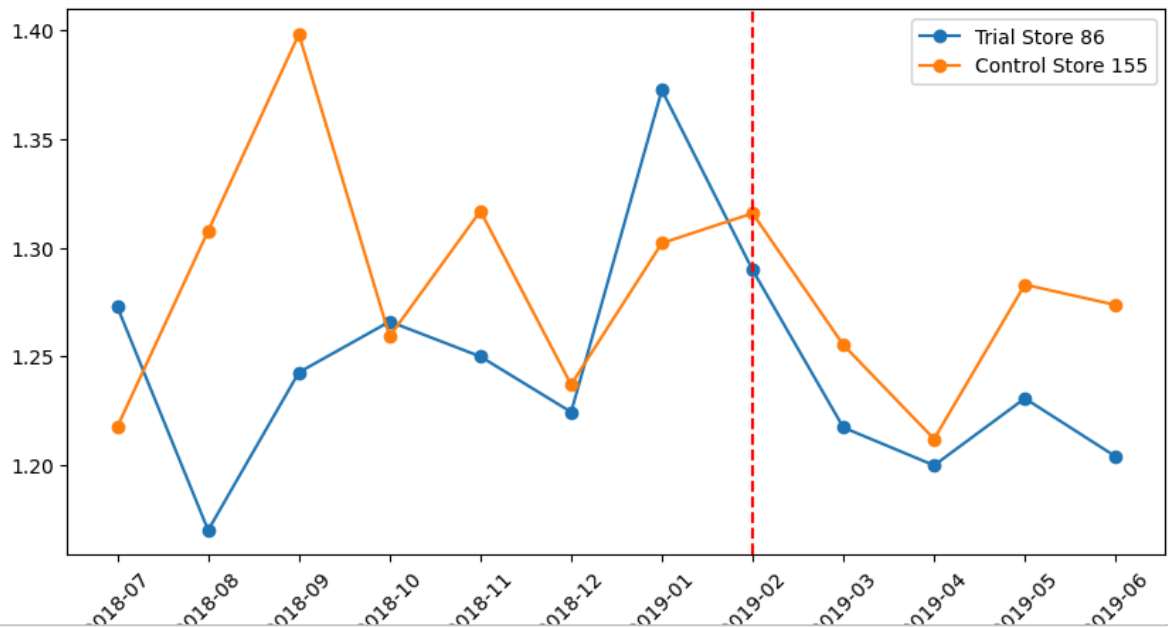
plt.plot(
    compare_86["YEAR_MONTH"].astype(str),
    compare_86["TXN_PER_CUSTOMER_CONTROL"],
    marker="o",
    label="Control Store 155"
)

plt.axvline("2019-02",color="red",linestyle="--")

plt.legend()

plt.xticks(rotation=45)

plt.show()
```



```
#graphs
#sales graph for store 88
plt.figure(figsize=(10,5))

plt.plot(
    compare_88["YEAR_MONTH"].astype(str),
    compare_88["TOTAL_SALES_TRIAL"],
    marker="o",
    label="Trial Store 88"
)

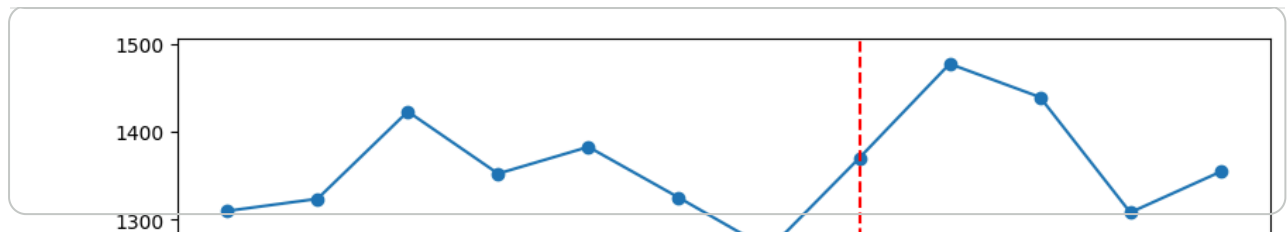
plt.plot(
    compare_88["YEAR_MONTH"].astype(str),
    compare_88["TOTAL_SALES_CONTROL"],
    marker="o",
    label="Control Store 125"
)

plt.axvline("2019-02",color="red",linestyle="--")

plt.legend()

plt.xticks(rotation=45)

plt.show()
```



#Trial Store 88 experienced higher sales during the trial period compared to

```
#graphs
#customers graph for store 88
plt.figure(figsize=(10,5))

plt.plot(
    compare_88["YEAR_MONTH"].astype(str),
    compare_88["TOTAL_CUSTOMERS_TRIAL"],
    marker="o",
    label="Trial Store 88"
```